

NETFORENSICS AI

AN INTELLIGENT NETWORK FAULT DETECTION AND TROUBLESHOOTING SYSTEM

PROJECT SUBMISSION REPORT

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NETFORENSICS AI - An Intelligent Network Fault Detection and Troubleshooting System

ABSTRACT

Computer networks are an essential part of modern organizations, educational institutions, industries, and digital services. When a network problem occurs, identifying its actual cause can be difficult, especially for beginner or junior network engineers. A connectivity problem may occur due to incorrect VLAN configuration, IP addressing issues, DHCP failure, DNS problems, routing errors, access control configurations, or other network-related faults. Although engineers may know individual networking commands, connecting network symptoms with the actual root cause often requires experience and systematic troubleshooting.

This project, **NetForensics AI – An Intelligent Network Fault Detection and Troubleshooting System**, is developed to assist in identifying and troubleshooting network faults in Cisco Packet Tracer-based environments. The system analyzes network symptoms and command outputs to provide a possible root cause, confidence level, evidence used, OSI layer, recommended next command, and suggested fix steps.

The project combines Artificial Intelligence assistance with deterministic rule-based validation and mandatory human review. The AI is not designed to independently apply network changes. Instead, every diagnosis requires review by a human before a fix is accepted or applied. This approach supports responsible use of Artificial Intelligence in network troubleshooting.

A structured dataset of network fault cases was prepared and cleaned for analysis. AI-assisted diagnosis was performed using structured prompts. A Python-based rule checker was implemented to identify common configuration mistakes. Human review results were recorded to compare AI recommendations with final human decisions. A dashboard was also created to visualize fault types, severity, AI confidence, diagnosis results, and human agreement.

Additionally, practical Cisco Packet Tracer test cases were developed to demonstrate the complete troubleshooting workflow. Each case follows a systematic process consisting of a working network, intentional fault generation, evidence collection, AI diagnosis, human review, fault correction, and final verification.

The project demonstrates that AI can act as an intelligent assistant for network troubleshooting while maintaining human control over important technical decisions.

Keywords: Artificial Intelligence, Network Troubleshooting, Cisco Packet Tracer, VLAN, DHCP, DNS, Routing, Human Review, Responsible AI, Network Fault Detection.

1. INTRODUCTION

Computer networks allow devices to communicate and exchange information. In a typical network environment, multiple devices such as computers, switches, routers, servers, and wireless devices work together. For communication to occur successfully, these devices must be correctly configured.

Even a small configuration mistake can result in communication failure. For example, a computer may be connected to the wrong VLAN, configured with an incorrect default gateway, unable to obtain an IP address from DHCP, unable to resolve a domain name because of DNS configuration, or unable to reach a remote network because of a missing route.

Traditional network troubleshooting requires engineers to collect symptoms, execute diagnostic commands, analyze command outputs, identify the root cause, and apply corrective actions. This process can be time-consuming and challenging for beginners.

Artificial Intelligence can assist in this process by analyzing available network evidence and suggesting possible causes and troubleshooting steps. However, AI-generated responses should not always be blindly trusted. AI may have incomplete information or may generate an incorrect diagnosis. Therefore, human oversight is important.

NetForensics AI was developed to address this requirement. The system acts as an AI-assisted network troubleshooting helper rather than an autonomous network administrator. It analyzes network symptoms and available evidence and provides a structured diagnosis. The diagnosis is then reviewed by a human before any corrective action is accepted.

The project combines AI assistance, rule-based validation, data analysis, visualization, and practical network simulations to create a complete intelligent troubleshooting workflow.

2. PROBLEM STATEMENT

Junior network engineers often understand individual networking commands but may face difficulty identifying the actual root cause of a network problem.

For example, when a computer receives an IP address but cannot communicate with a server, several possible causes may exist. The problem may be related to VLAN configuration, routing, DNS, DHCP, gateway configuration, or another network component.

The main challenge is not only identifying that a problem exists but determining:

- What is the likely root cause?
- Which OSI layer is affected?
- What evidence supports the diagnosis?
- Which command should be executed next?
- What configuration change may solve the problem?

Traditional troubleshooting depends heavily on technical knowledge and experience.

Therefore, there is a need for an intelligent system that can analyze network symptoms and command outputs, provide structured troubleshooting assistance, and maintain human control over the final decision.

NetForensics AI addresses this problem by providing an AI-assisted troubleshooting workflow supported by rule-based validation and mandatory human review.

3. PROJECT OBJECTIVES

The major objectives of NetForensics AI are as follows:

1. To create an intelligent network fault detection and troubleshooting system.
2. To prepare a structured dataset containing multiple network troubleshooting cases.
3. To analyze network symptoms and Cisco-style command outputs using AI.
4. To identify the likely root cause of a network problem.
5. To determine the possible OSI layer associated with the fault.
6. To provide a confidence level for AI-generated diagnoses.
7. To recommend the next troubleshooting command.
8. To provide evidence-based suggested fix steps.
9. To implement rule-based validation for common network configuration mistakes.
10. To ensure human review before accepting an AI-generated diagnosis.
11. To maintain responsible AI records where human reviewers accept, edit, or reject AI responses.
12. To visualize network fault patterns and AI performance using a dashboard.
13. To demonstrate practical troubleshooting cases using Cisco Packet Tracer.

4. SCOPE OF THE PROJECT

The scope of NetForensics AI is focused on troubleshooting common networking faults in simulated Cisco Packet Tracer environments.

The project covers areas such as:

- VLAN-related faults
- DHCP issues
- DNS configuration problems
- Static routing failures
- Default gateway issues
- IP addressing problems
- Interface configuration errors
- Network connectivity failures

The system uses network symptoms and command outputs as evidence for diagnosis.

The project does not automatically change network configurations. Instead, it follows a human-in-the-loop approach where AI suggestions must be reviewed before applying a solution.

The system is intended primarily as an educational and decision-support tool for network troubleshooting.

5. PROPOSED SOLUTION

The proposed solution is **NetForensics AI – An Intelligent Network Fault Detection and Troubleshooting System**.

NetForensics AI combines the following components:

- Network fault case dataset
- Data cleaning and preparation
- AI-assisted diagnosis
- Structured prompt engineering
- Rule-based fault detection
- Human review
- Responsible AI logging
- Data visualization dashboard
- Cisco Packet Tracer testing

The system receives information about a network fault, such as symptoms and terminal output. The available evidence is analyzed using a structured AI prompt.

The AI provides information including:

- Likely root cause
- Confidence level
- Evidence used
- OSI layer
- Recommended next command
- Suggested fix steps

The response is then reviewed by a human. The reviewer can mark the AI diagnosis as:

- Accepted
- Edited
- Rejected

After human approval, the recommended fix can be applied to the network and the connectivity can be verified.

This approach ensures that AI acts as an assistant rather than replacing human technical judgment.

6. SYSTEM WORKFLOW

The overall workflow of NetForensics AI follows a systematic process.

Step 1: Network Fault Creation

A network topology is created using Cisco Packet Tracer. The network is first configured correctly to establish a working baseline.

Step 2: Working Verification

Connectivity between devices is tested to confirm that the network is functioning correctly.

Step 3: Fault Generation

An intentional configuration fault is introduced into the network.

Step 4: Symptom Observation

The effect of the fault is observed through failed connectivity or unexpected network behavior.

Step 5: Evidence Collection

Relevant command outputs or terminal results are collected from the affected device.

Examples include:

- ipconfig
- ping
- show vlan brief
- show interfaces
- show ip route

Step 6: AI-Assisted Analysis

The collected evidence is provided to NetForensics AI.

Step 7: AI Diagnosis

The AI generates a structured response containing the likely cause and troubleshooting recommendations.

Step 8: Human Review

A human reviewer evaluates the AI diagnosis and marks it as Accepted, Edited, or Rejected.

Step 9: Apply Approved Fix

The approved configuration correction is applied to the network.

Step 10: Verification

Connectivity is tested again to confirm that the problem has been resolved.

7. SYSTEM ARCHITECTURE

The NetForensics AI system consists of multiple interconnected components.

The process begins with a simulated network environment created using Cisco Packet Tracer. When a network fault occurs, relevant symptoms and command outputs are collected.

The collected evidence is passed to the AI-assisted diagnosis component through structured prompts.

The AI diagnosis is supported by a rule-based network validation component. The rule checker performs deterministic checks for common configuration mistakes.

The AI-generated response is then passed to the human review stage. The reviewer evaluates the response and decides whether it should be accepted, edited, or rejected.

The results of AI diagnosis and human review are stored in CSV files for further analysis.

Finally, a dashboard visualizes network fault types, severity levels, AI confidence, and human agreement.

The overall architecture can be represented as:

Cisco Packet Tracer Network
↓
Fault Generation
↓
Evidence Collection
↓
NetForensics AI Analysis
↓
Rule-Based Validation
↓
Human Review
↓
Approved Fix
↓
Network Verification
↓
Results and Dashboard

8. DATASET PREPARATION AND VALIDATION

The development of NetForensics AI required a structured collection of network troubleshooting cases that could be used for analysis, AI-assisted diagnosis, rule-based validation, and human review. The original network fault cases dataset was used as the primary source of information throughout the project.



Case_ID	Category	Difficulty	Topology	Symptom	Evidence	Expected_Fault	OSI_Layer	Concept	Severity	Expected_Next_Comm	Expected_Fix	Verification	Notes
C001	VLAN	Easy	PC1->Switch1->Route	PC cannot reach gate	show vlan brief, show	PC switch port assign	Layer 2	VLAN	High	show vlan brief	Assign Fa0/1 to VLAN	Ping gateway and ser	PC has VLAN 10 IP configuration but switch
C002	VLAN	Easy	PC1->Switch1->Route	PC cannot reach gate	VLAN and switchport	VLAN 10 missing on S	Layer 2	VLAN	High	show vlan brief	Recreate VLAN 10 and	Gateway and server p	VLAN itself was deleted
C003	VLAN/Trunk	Medium	PC1->Switch1->Switch	PC cannot reach PC2	show interfaces trunk	VLAN 10 not allowed	Layer 2	Trunking	High	show interfaces trunk	Allow VLAN 10 on tru	PC1 successfully ping	VLAN 10 exists on both switches and access
C004	DHCP	Medium	PC1-> Switch1-> Rout	PC1 does not receive	ipconfig, show ip dhc	DHCP pool configured	Layer 3	DHCP	High	show ip dhcp pool	Configure DHCP pool	PC1 receives 192.168	DHCP service is present, but the pool netwo
C005	Interface Failure	Easy	PC1->Switch1->Route	PC1 cannot reach defi	show ip interface brie	R1 G0/0 administrati	Layer 3	Router interface statu	High	show ip interface brie	no shutdown on Rout	G0/0 becomes up/up	Physical connection is intact, but the router i
C006	IP Addressing	Medium	PC1-> Switch1-> Rout	Connectivity fails afte	ipconfig, arp -a on PC	PC1 and PC2 configur	Layer 3	Duplicate IP Addressi	High	ipconfig	Assign PC2 a unique I	PC1 and PC2 successf	Duplicate IP addresses cause an IP address
C007	Routing	Medium	PC1->Switch1->Route	PC1 cannot reach PC2	show ip route, show ir	R1 has incorrect static	Layer 3	Static Routing	High	show ip route	Change next hop to 1	PC1 successfully ping	Remote network exists, but R1 forwards traf
C008	DNS/Name Resolution	Easy	PC1->Switch1->Route	IP connectivity works	ipconfig, successful D	PC1 configured with i	Layer 7	DNS	Medium	ipconfig/nslookup	Set DNS server to 192	nslookup cisco.local su	Network connectivity remains available; only
C009	Ethernet	Easy	PC1->Switch1->PC2	PC1 cannot communic	show interfaces status	Ethernet cable betwe	Layer 1	Physical connectivity/	High	show interfaces status	Reconnect the Ethern	Ping PC2 from PC1 op	Physical link should show connected/green
C010	Ethernet	Easy	PC1->Switch1	PC1 cannot communic	show interfaces status	Wrong Ethernet cable	Layer 1	Ethernet cable type/s	High	show interfaces status	Replace the incorrect	show interfaces status	Wrong cable type used; physical link is down
C011	ARP / Address Resolu	Medium	PC1 → Router1 : Rout	PC1 cannot reach PC2	show ip interface brie	Incorrect static ARP e	Layer 2/3	ARP address-to-MAC r	High	show arp	Remove the incorrect	ping 192.168.20.10 su	Incorrect MAC mapped to PC2 IP
C012	TCP / HTTP Service	Medium	PC1 → Switch1 : Switc	PC1 can ping the serv	Ping to 192.168.10.2(	HTTP service disablec	Layer 4/7	HTTP service over TC	Medium	ping 192.168.10.20	Enable HTTP service (	HTTP webpage loads	IP connectivity works but HTTP service is un
C013	CDP / Neighbor Disco	Medium	PC1 → Router1 : Rou	R1 cannot discover R2	show cdp neighbors d	CDP disabled on R1 G	Layer 2	Cisco Discovery Protoc	Medium	show cdp neighbors	Enable CDP on R1 G0	R2 appears in show c	CDP disabled on R1 G0/1 while IP connectiv
C014	STP	Medium	Switch1 → Switch2-Sv	SW3 unexpectedly be	show spanning-tree a	Incorrect STP priority	Layer 2	STP Root Bridge Electi	Medium	show spanning-tree	Set SW3 VLAN 1 prior	show spanning-tree v	SW3 priority was set to 0, causing it to win tl
C015	EtherChannel / PAgP	Medium-Hard	Switch1 → Switch2 : E	EtherChannel is not o	show etherchannel su	Switch2 assigned to tl	Layer 2	Etherchannel / PAgP	High	show etherchannel su	Remove Fa0/2 from c	show etherchannel su	One physical member was assigned to a diff
C016	Network Discovery	Medium	Router1 ↔ Router2 :	R1/R2 cannot discove	show cdp neighbors, s	CDP disabled on R1 G	Layer 2	Cisco Discovery Protoc	Medium	show cdp neighbors d	Enable CDP on R1 G0	show cdp neighbors s	IP connectivity is working, but CDP neighbor
C017	DHCP Services	Medium	PC1 → Switch1 ↔ Rou	PC1 displays "DHCP n	PC DHCP request faile	Missing ip helper-add	Layer 3	DHCP Relay	High	show running-config	ip helper-address on t	PC receives a valid DH	Verify end-to-end connectivity first. Capture
C018	Switching	Medium	PC1 → Switch1 ↔ Swi	The trunk is operation	show interfaces trunk	Different native VLAN	Layer 2	Native VLAN	Medium	show interfaces trunk	Configure the same n	Verify trunk and succe	Compare trunk settings on both ends. Recon
C019	Switching	Medium	Switch1 ↔ Switch2 wi	Hosts in the same VLA	show interfaces trunk	Incorrect DTP negotia	Layer 2	Dynamic Trunking Pro	Medium	show interfaces switcl	Configure compatible	Trunk becomes active	Check the trunk allowed-VLAN list before ch
C020	Switching	Medium	Switch1-Switch2-PCs	A host cannot commu	show vtp status, show	Incorrect VTP domain	Layer 2	VTP	High	show vtp status	Set the correct VTP dc	Verify VLAN database	Confirm intended VLAN and IP network. Rec
C021	Switching	Medium	Switch1-Switch2-PCs	VLAN information cre	show vtp status, show	VTP server/client mo	Layer 2	VTP	High	show vtp status	Correct VTP operating	Verify VLAN appears	Compare VTP domain, mode, and revision in
C022	Switching	Medium	Switch1-Switch2-Swi	Connectivity works, bu	show spanning-tree	Incorrect STP root bri	Layer 2	STP Root Bridge	Medium	show spanning-tree	Set intended root prio	Verify root bridge and	Document the current root bridge and port
C023	Switching Security	Hard	Access switch with en	An access interface be	show spanning-tree, s	PortFast configured o	Layer 2	STP PortFast	High	show spanning-tree	Apply PortFast only to	Verify STP state	Capture interface and STP evidence before
C024	Switch Security	Hard	SW1 with access PC	A switch port restricts	show interfaces status	BPDU Guard triggerec	Layer 2	BPDU Guard	High	show interfaces status	Remove unintended E	Verify interface is con	Record port-security status, violation count,
C025	Switch Security	Hard	SW1-SW2-SW3 train	A replacement device	show spanning-tree d	Root Guard is blocking	Layer 2	Root Guard	High	show spanning-tree	Correct/remove Root	Verify port returns to	Collect the existing sticky MAC entry. Remov
C026	Switching	Medium	PC1-SW1 access port	EtherChannel forms s	show port-security, sh	Port-security maximu	Layer 2	Port Security	High	show port-security	Set appropriate maxir	Verify secure port anc	Compare channel members: switchport mod
C027	Switching	Medium	PC1-SW1 access port	EtherChannel negotia	show port-security ad	Sticky MAC violation	Layer 2	Sticky MAC	High	show port-security ad	Clear/update sticky M	Verify correct secure i	Check channel summary on both sides. Conf
C028	Switching	Medium	PC1-SW1 access port	A device MAC address	show port-security	Wrong violation mode	Layer 2	Port Security Violation	Medium	show port-security	Configure intended vi	Reconnect/test endpo	Generate traffic and capture the MAC table.
C029	Routing	Medium	PC1-SW1-PC2	Local connectivity wor	show mac address-tal	Incorrect static MAC e	Layer 2	MAC Address Table	Medium	show mac address-tal	Correct/remove the st	Verify MAC is learned	Record the routing table before fixing. Add c
C030	Routing	Medium	SW1 with multiple PC	A static route exists bu	show mac address-tal	Incorrect physical con	Layer 2	CAM/MAC Learning	Medium	show mac address-tal	Correct cabling/port-s	Verify MAC appears o	Validate next-hop reachability. Capture the
C031	Routing	Medium	LAN1-R1-R2-LAN2	Traffic uses the backu	show ip route; show ij	Missing default route	Layer 3	Default Routing	High	show ip route	Configure correct defi	Ping remote network	Document both routes and AD values. Corre
C032	Dynamic Routing	Medium	R1 with primary and l	A remote network is n	show ip route; show r	Incorrect administrati	Layer 3	Administrative Distan	Medium	show ip route	Correct static-route ac	Verify preferred routi	Check RIP participating networks. Add the r
C033	Dynamic Routing	Medium	LAN1-R1-R2-LAN2	RIP routers do not exc	show ip route; show r	Incorrect static-route	Layer 3	Static Routing	High	show ip route	Correct next-hop addi	Ping destination and	Capture protocol settings on both routers. St
C034	Dynamic Routing	Medium	R1-R2-R3	EIGRP neighbors do n	show ip protocols; sho	Missing RIP network s	Layer 3	RIP v2	High	show ip protocols	Advertise the missing	Verify RIP route appe	Verify basic connectivity, compare EIGRP AS
C035	Dynamic Routing	Medium	R1-R2-R3 with multi	EIGRP adjacency does	show ip protocols; sho	RIP auto-summary ca	Layer 3	RIP Auto-Summary	High	show ip protocols	Disable auto-summar	Verify specific routes i	Identify the passive interface. Remove passi

Figure 8.1 Original Network Fault Cases dataset

The dataset contains different networking fault scenarios representing common problems that may occur in computer networks. Each case provides relevant information about a network issue, allowing the system to perform structured troubleshooting analysis. The cases cover multiple networking concepts and technologies, helping ensure that the system is capable of analyzing different types of network faults.

Before using the dataset for further analysis, data preparation and validation activities were performed using Python and Jupyter Notebook. The purpose of this process was to examine the dataset structure and ensure that the required information was available in a suitable format for processing.

The validation process focused on reviewing the available columns, checking the completeness of important fields, identifying missing or inconsistent values, and ensuring that the case information could be effectively used by different components of the system. This step was important because the quality and reliability of AI-assisted analysis depend significantly on the quality of the input data.

The dataset was examined to ensure that each network fault case contained meaningful information related to the troubleshooting process. Important information included network symptoms, protocols or technologies involved, possible faults, severity levels, and other relevant troubleshooting details.

The data preparation and validation stage included reviewing the overall dataset structure, checking the available columns and their relevance, examining network fault case descriptions, checking for missing or incomplete information, reviewing data consistency across different cases, and validating the fields required for further analysis.

It is important to note that a separate cleaned dataset file was not created for this project. Instead, the original network fault cases dataset was validated and used directly as the primary input for the subsequent stages of the NetForensics AI workflow.

After validation, the dataset was processed through multiple stages. The first stage involved AI-assisted diagnosis, where structured prompts were used to analyze network fault cases and generate possible diagnoses. The AI-generated results were stored separately in the AI-Assisted Diagnosis Results file.

```
print("Dataset Columns:")

for column in df.columns:
    print(column)
[9]
```

```
Dataset Columns:
Case_ID
Category
Difficulty
Topology
Symptom
Evidence
Expected_Fault
OSI_Layer
Concept
Severity
Expected_Next_Command
Expected_Fix
Verification
Notes
```

Figure 8.2 Dataset Overview

```
missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)
[11]
```

```
Missing values in each column:
Case_ID                  0
Category                 0
Difficulty               0
Topology                 0
Symptom                  0
Evidence                 0
Expected_Fault           0
OSI_Layer                0
Concept                  0
Severity                 0
Expected_Next_Command    0
Expected_Fix             0
Verification             0
Notes                    0
```

Figure 8.3 Missing Value Validation

The network cases were also analyzed using deterministic rule-based checks to identify common configuration mistakes and network faults. The results generated from this process were stored in the Rule-Based Detection Results file.

Following the AI-assisted analysis, the generated diagnoses were evaluated through a human-in-the-loop process. Human reviewers examined the AI responses and recorded whether each diagnosis was accepted, edited, or rejected. These decisions were maintained separately in the Human Review Results file.

Therefore, the dataset preparation and validation process served as the foundation for the complete NetForensics AI workflow. The validated original dataset was used as the main input, while separate result files were generated for AI-assisted diagnosis, rule-based detection, and human review.

This approach helped maintain a clear separation between the original network fault cases and the results generated during different stages of the project. The original dataset remained the primary source of network scenarios, while the generated outputs were used for comparison, evaluation, and visualization.

The overall data flow of the project follows the sequence of original network fault cases, data preparation and validation, AI-assisted diagnosis, rule-based fault detection, human review, results and final analysis, and dashboard visualization.

9. AI PROMPT ENGINEERING

Prompt engineering plays an important role in the NetForensics AI system.

Instead of asking the AI a general question, structured prompts were designed to guide the model toward a consistent troubleshooting response.

The main prompt instructs the AI to analyze the available network information without making unsupported assumptions.

The AI is requested to provide results in a structured format containing:

- Root Cause
- Confidence
- Evidence Used
- OSI Layer
- Recommended Next Command
- Suggested Fix Steps

The prompt also emphasizes that insufficient evidence should be clearly identified.

An important instruction included in the prompt is:

"Do not make unsupported assumptions."

This is important because network troubleshooting often requires multiple pieces of evidence. If only limited information is available, the AI should recommend further investigation rather than confidently guessing the cause.

The prompt library was maintained using Markdown files to organize and document the AI instructions.

The use of structured prompts improves:

- Consistency
- Explainability
- Evidence-based reasoning
- Response quality
- Human review

diagnose_prompt.md

C: > Users > Urja Chauhan > OneDrive > 文档 > NetForensics AI > 04_Prompts > diagnose_prompt.md > #### NetForensics AI: An Intelligent Network Fault Detection and Troubleshooting System

```
#### **NetForensics AI: An Intelligent Network Fault Detection and Troubleshooting System**

######

##### Diagnosis Prompt Template



###### Purpose



This prompt is used by NetForensics AI to analyze network fault cases and generate an AI-assisted troubleshooting diagnosis.



The AI analyzes the available network topology, protocol or technology, symptoms, and command evidence. The diagnosis is generated based only on the provided in
reviewed by a human reviewer.







###### Prompt



You are a network troubleshooting assistant.



Analyze the following network fault case using only the information provided.



Network Topology:

{topology}



Protocol or Technology:

{Category}



Symptom:

{symptom}
```

Figure 9.1 Main Structured Diagnosis Prompt Used by NetForensics AI

10. AI-ASSISTED NETWORK FAULT DIAGNOSIS

The AI-assisted diagnosis component analyzes network troubleshooting cases using the structured prompt library.

For each case, available information is provided to the AI.

The AI attempts to identify the most likely root cause based on the evidence.

A typical response includes:

Root Cause

The most likely technical reason for the observed network problem.

Confidence

A confidence level indicating how certain the AI is based on the available evidence.

Evidence Used

The information or command output used to support the diagnosis.

OSI Layer

The network layer most closely associated with the fault.

Recommended Next Command

A suggested command that may provide additional troubleshooting information.

Suggested Fix Steps

Recommended actions for resolving the identified problem.

The AI output is stored for comparison and analysis.

The system does not assume that the AI is always correct. Every AI-generated diagnosis is subject to human review.

```python
example_diagnosis = {
    "Root Cause": "Possible network configuration issue",
    "Confidence": "Medium",
    "Evidence Used": "Network connectivity failure observed",
    "OSI Layer": "Network Layer",
    "Recommended Next Command": "show ip interface brief",
    "Suggested Fix Steps": "Verify interface status and IP configuration"
}

for key, value in example_diagnosis.items():
    print(key + ":", value)
```

[11]

```
Root Cause: Possible network configuration issue
Confidence: Medium
Evidence Used: Network connectivity failure observed
OSI Layer: Network Layer
Recommended Next Command: show ip interface brief
Suggested Fix Steps: Verify interface status and IP configuration
```

Figure 10.1 AI-Assisted Network Fault Diagnosis Generated by NetForensics AI

11. RULE-BASED NETWORK FAULT DETECTION

In addition to AI-based analysis, a rule-based network fault detection component was implemented.

The purpose of this component is to perform deterministic checks for common and easily verifiable network configuration problems.

Unlike AI, rule-based checks follow predefined conditions.

The system can identify common issues such as:

- Duplicate IP addresses
- Incorrect subnet masks
- Gateway mismatch
- Interface down status
- Missing VLAN configuration
- Missing routes

For example, if two devices are configured with the same IP address, the rule checker can identify this problem based on a direct comparison.

Similarly, if a routing table does not contain a required network route, the system can detect the missing route.

The rule-based component is useful because some network problems can be detected with clear logical rules without requiring AI interpretation.

The combination of AI and deterministic checks provides a more reliable troubleshooting approach.

```
df[[
    "Case_ID",
    "Expected_Fault",
    "Rule Detected Fault",
    "Rule Recommended Command",
    "Rule Suggested Fix"
]].head(10)
[10]
```

	Case_ID	Expected_Fault	Rule Detected Fault	Rule Recommended Command	Rule Suggested Fix
0	C001	PC switch port assigned to wrong VLAN	Unknown Fault	Check network configuration	Manual investigation required
1	C002	VLAN 10 missing on Switch1	Unknown Fault	Check network configuration	Manual investigation required
2	C003	VLAN 10 not allowed on trunk	Unknown Fault	Check network configuration	Manual investigation required
3	C004	DHCP pool configured with wrong network	DHCP Configuration Issue	ipconfig /all	Check the DHCP configuration
4	C005	R1 G0/0 administratively shut down	Unknown Fault	Check network configuration	Manual investigation required
5	C006	PC1 and PC2 configured with the same IP addres...	Unknown Fault	Check network configuration	Manual investigation required
6	C007	R1 has incorrect static route next hop 10.0.0.3	Unknown Fault	Check network configuration	Manual investigation required
7	C008	PC1 configured with incorrect DNS server 192.1...	DNS Configuration Issue	nslookup	Check the DNS server configuration
8	C009	Ethernet cable between Switch1 Fa0/2 and PC2 F...	Connectivity Issue	ping	Check IP addressing and network connectivity
9	C010	Wrong Ethernet cable type used between PC1 Fa0...	Unknown Fault	Check network configuration	Manual investigation required

Figure 11.1 Rule-Based Network Fault Detection Results Generated by NetForenics AI

12. HUMAN REVIEW AND RESPONSIBLE AI

Responsible AI is a major component of the NetForensics AI project.

AI-generated answers may be useful, but they should not automatically be considered correct.

A human reviewer is therefore included in the troubleshooting workflow.

The reviewer evaluates each AI-generated diagnosis and selects one of the following decisions:

Accepted

The AI diagnosis is correct and supported by the available evidence.

Edited

The AI diagnosis is partially correct, but the human reviewer modifies the diagnosis or suggested solution.

Rejected

The AI diagnosis is incorrect or unsupported by the available evidence.

This process ensures that final network decisions remain under human control.

Human review is particularly important when:

- Evidence is incomplete
- Multiple faults are possible
- AI confidence is low
- The suggested fix may affect network configuration
- The AI makes an unsupported assumption

The human review results are stored separately for analysis.

The project also maintains responsible AI examples where the AI response required human correction or additional investigation.

This demonstrates that the purpose of NetForensics AI is not to replace network engineers but to support them during troubleshooting.

```
df[[
    "Case_ID",
    "Symptom",
    "Evidence",
    "Expected_Fault",
    "AI_Probable_Fault",
    "AI_Confidence",
    "Human_Review_Status",
    "Human_Review_Priority",
    "Human_Reviewer_Decision"
]].head(10)
```
[16]

	Case_ID	Symptom	Evidence	Expected_Fault	AI_Probable_Fault	AI_Confidence	Human_Review_Status	Human_Review_Priority	Human_Reviewer_Decision
0	C001	PC cannot reach gateway or server	show vlan brief, show interfaces fa0/1 switchp...	PC switch port assigned to wrong VLAN	VLAN 10 missing on Switch1	Medium	Human Verification Recommended	Medium Priority	Pending
1	C002	PC cannot reach gateway/server	VLAN and switchport outputs + IP/ping results	VLAN 10 missing on Switch1	PC switch port assigned to wrong VLAN	Medium	Human Verification Recommended	Medium Priority	Pending
2	C003	PC1 cannot reach PC2	show interfaces trunk on Switch1/Switch2; show...	VLAN 10 not allowed on trunk	R1 has incorrect static route next hop 10.0.0.3	Medium	Human Verification Recommended	Medium Priority	Pending
3	C004	PC1 does not receive a vali 192.168.10.x address	ipconfig, show ip dhcp pool, show ip dhcp bind...	DHCP pool configured with wrong network	Missing ip helper-address	Medium	Human Verification Recommended	Medium Priority	Pending

Figure 12.1 Final Human Review and Responsible AI Decisions

13. NETFORENSICS AI APPLICATION

A user-facing application was developed for NetForensics AI.

The application provides an interface through which network troubleshooting information can be analyzed.

The user can provide relevant details about a network problem, such as:

- Network topology
- Protocol or technology
- Network symptom
- Terminal output
- Command evidence

The application processes the information using the AI-assisted diagnosis workflow.

The generated output provides a structured troubleshooting response.

The application demonstrates how the analytical components of the project can be transformed into a usable system.

The main purpose of the application is to make the troubleshooting process easier to understand and access.

Instead of manually analyzing multiple pieces of information, the user can provide available evidence to NetForensics AI and receive a structured recommendation.

However, the final decision remains subject to human review.

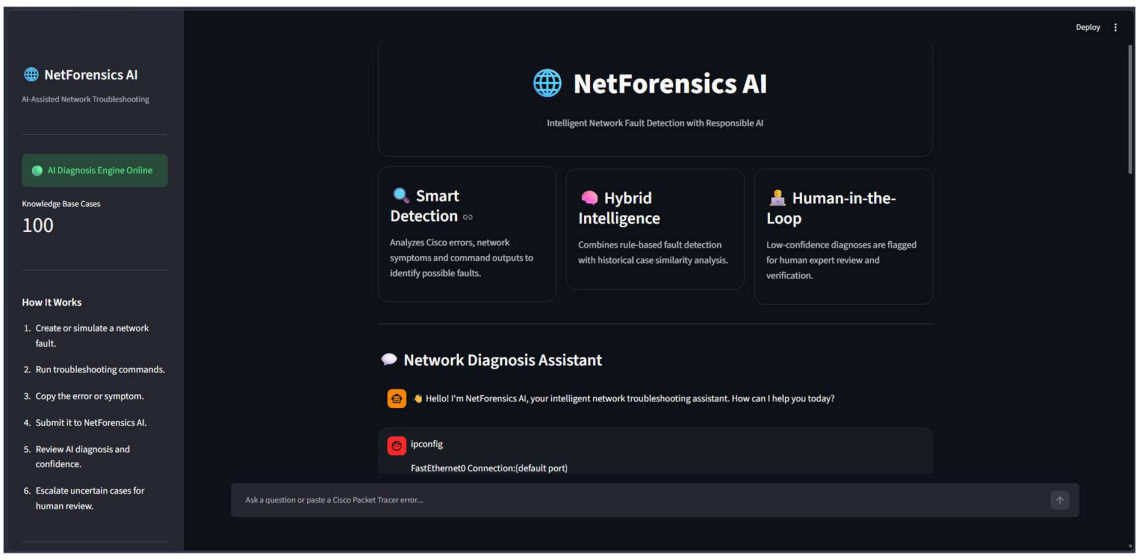


Figure 13.1 NetForensics AI Application Interface

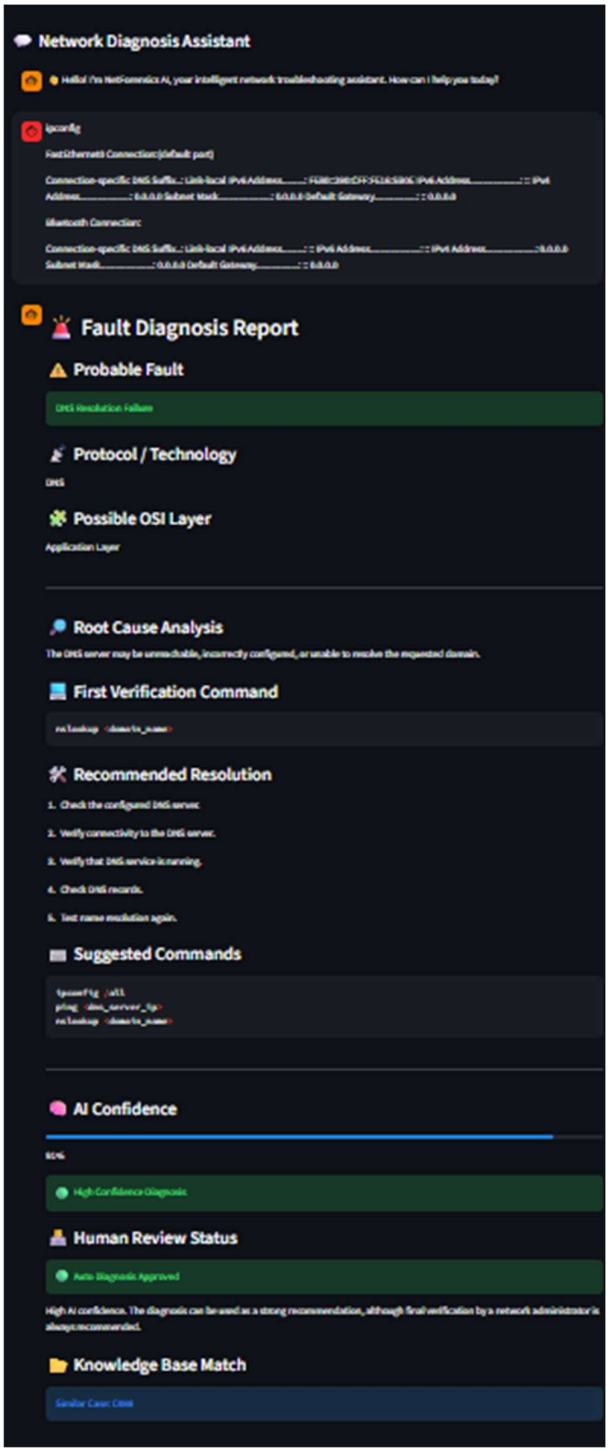


Figure 13.2 AI Diagnosis Result in Application

14. CISCO PACKET TRACER TEST CASE IMPLEMENTATION

Practical testing was performed using Cisco Packet Tracer.

For each test case, a working network topology was first created.

The process followed the sequence:

1. Create the network topology.
2. Configure devices correctly.
3. Verify successful communication.
4. Save the working network.
5. Create a copy of the working network.
6. Introduce a network fault.
7. Observe the resulting symptom.
8. Collect command output as evidence.
9. Analyze the evidence using NetForensics AI.
10. Perform human review.
11. Apply the approved solution.
12. Verify successful communication.

This methodology demonstrates the complete troubleshooting lifecycle.

The project includes multiple test cases involving different networking technologies.

14.1 Test Case 01 – VLAN Configuration Fault

The first test case demonstrates a VLAN-related connectivity problem.

A working network was initially configured where the device could communicate successfully.

A fault was introduced by assigning a switch port to an incorrect VLAN.

As a result, the affected device was unable to communicate with the expected network.

Relevant switch configuration output was collected as evidence.

NetForensics AI analyzed the evidence and identified the likely VLAN mismatch.

The diagnosis was reviewed by a human before applying the correction.

After assigning the switch port to the correct VLAN, connectivity was successfully restored.

CASE 01 – VLAN Connectivity

Objective:
Establish and verify normal VLAN communication.

Network Status:
PC1 can successfully communicate with its default gateway and Server1.

Verification:
Connectivity is confirmed before introducing the fault.

Next Stage:
A VLAN-related fault will be introduced for troubleshooting.



Figure 14.1.1 Working VLAN Topology

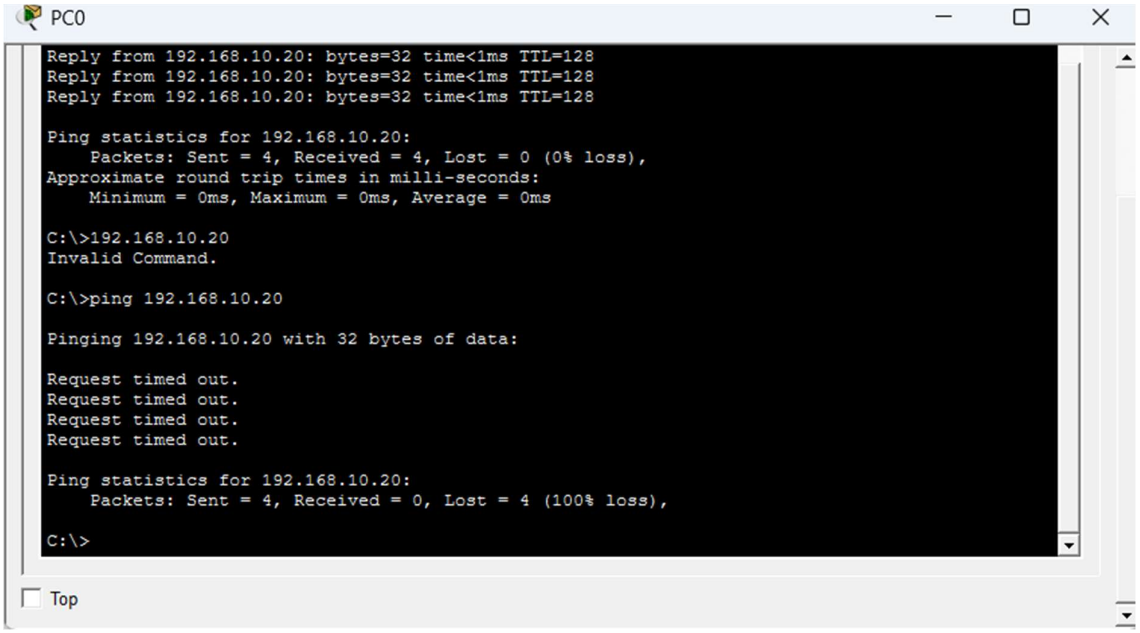


Figure 14.1.2 VLAN Fault Evidence

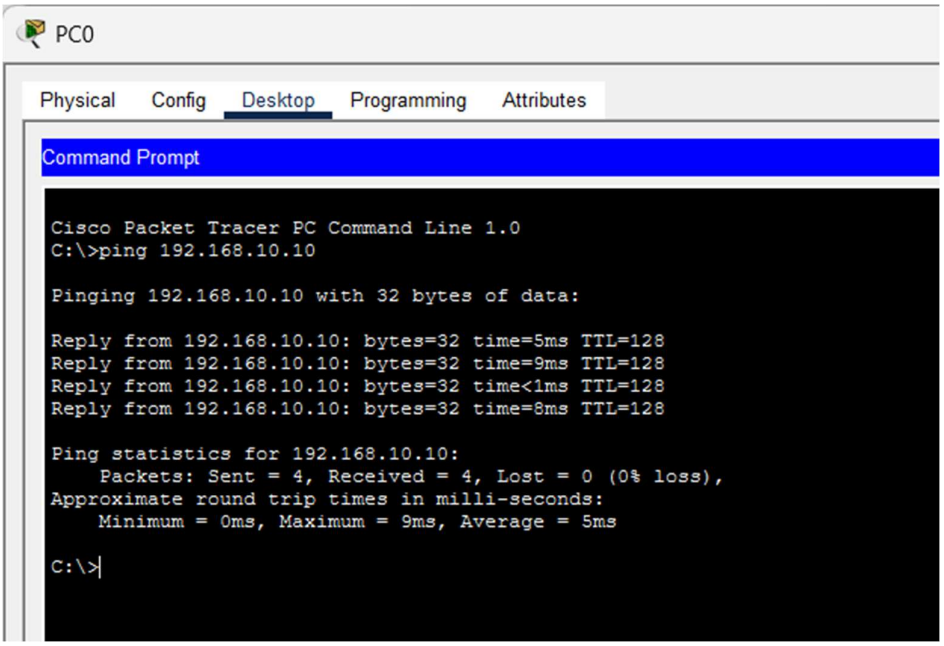


Figure 14.1.3 Successful Working

14.2 Test Case 02 – VLAN Communication Failure with Human Review

The second test case focuses on responsible AI and insufficient evidence.

Two computers were configured within the same IP network but were unable to communicate.

Initially, only limited evidence was provided to the AI.

Instead of making an unsupported assumption, the AI was expected to indicate uncertainty and recommend further investigation.

The human reviewer then collected additional switch configuration evidence.

The investigation revealed that the switch ports belonged to different VLANs.

The final diagnosis confirmed an incorrect VLAN configuration.

The affected port was assigned to the correct VLAN and communication was verified.

This case demonstrates the importance of human involvement when the available evidence is insufficient.

CASE 02 – VLAN Communication

Objective:
Demonstrate successful communication between two PCs configured in the same VLAN.

Initial Status:
PC0 and PC1 can successfully communicate with each other.

Verification:
Successful ping confirms the working network configuration.

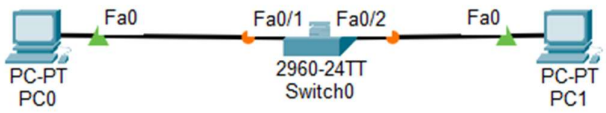


Figure 14.2.1 Failed Connectivity

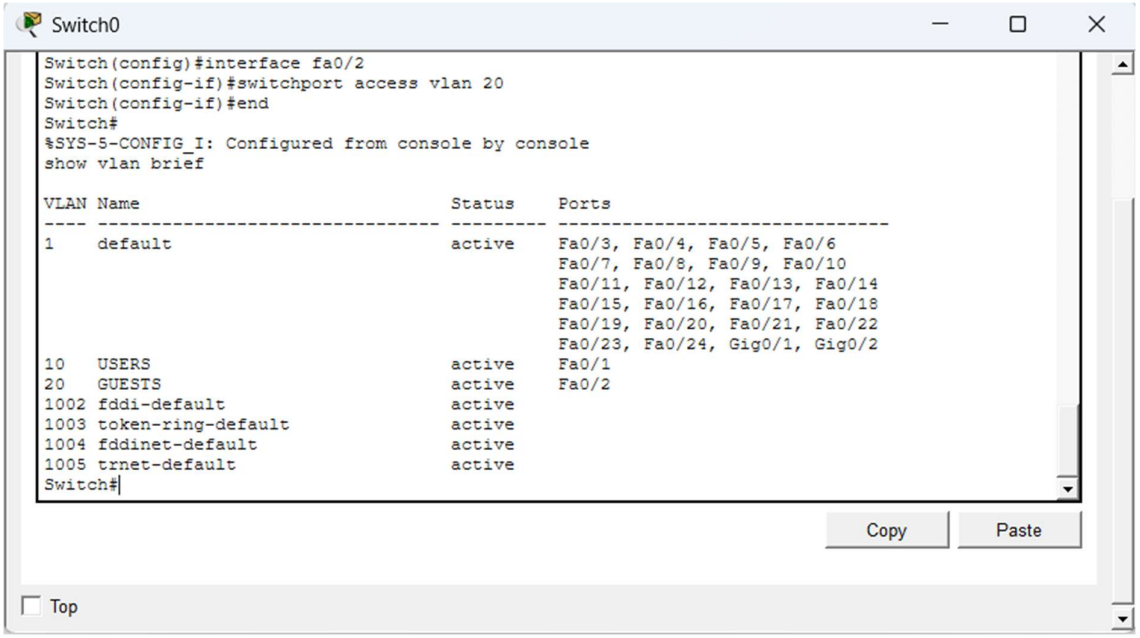


Figure 14.2.2 Human Investigation Evidence

CASE 02 – VLAN Communication Failure with Human Review

Objective:
Demonstrate Responsible AI by handling a network connectivity fault.

Initial Symptom:
PC0 cannot ping PC1 despite being configured in the same IP subnet.

AI Analysis:
The available evidence is analyzed by NetForensics AI before determining the root cause.

Human Review:
Human investigation validates or corrects the AI diagnosis.

Resolution:
Apply the approved configuration change and verify connectivity.

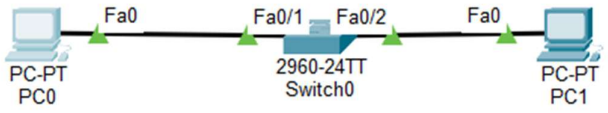


Figure 14.2.3 Successful connectivity after Fix

14.3 Test Case 03 – DHCP Address Assignment Failure

The third test case focuses on DHCP.

A DHCP server was configured to automatically assign IP addresses to client devices.

The working network was first verified by confirming that the client received a valid IP address.

A fault was then introduced by disabling or modifying the DHCP service.

The client was unable to receive a valid network configuration.

The client configuration output was collected as evidence.

NetForensics AI analyzed the available information and identified a possible DHCP service or configuration problem.

Human investigation confirmed the actual cause.

After restoring the DHCP configuration, the client successfully obtained an IP address.

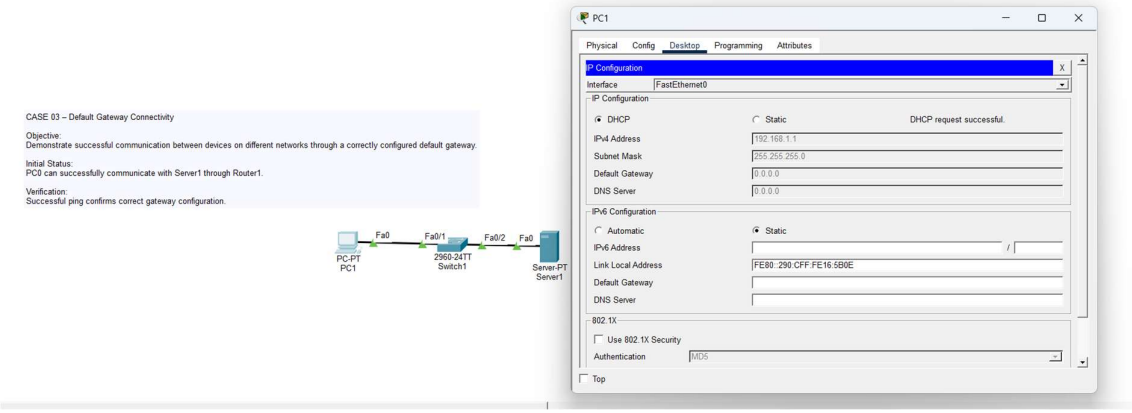


Figure 14.3.1 Working DHCP Assignment

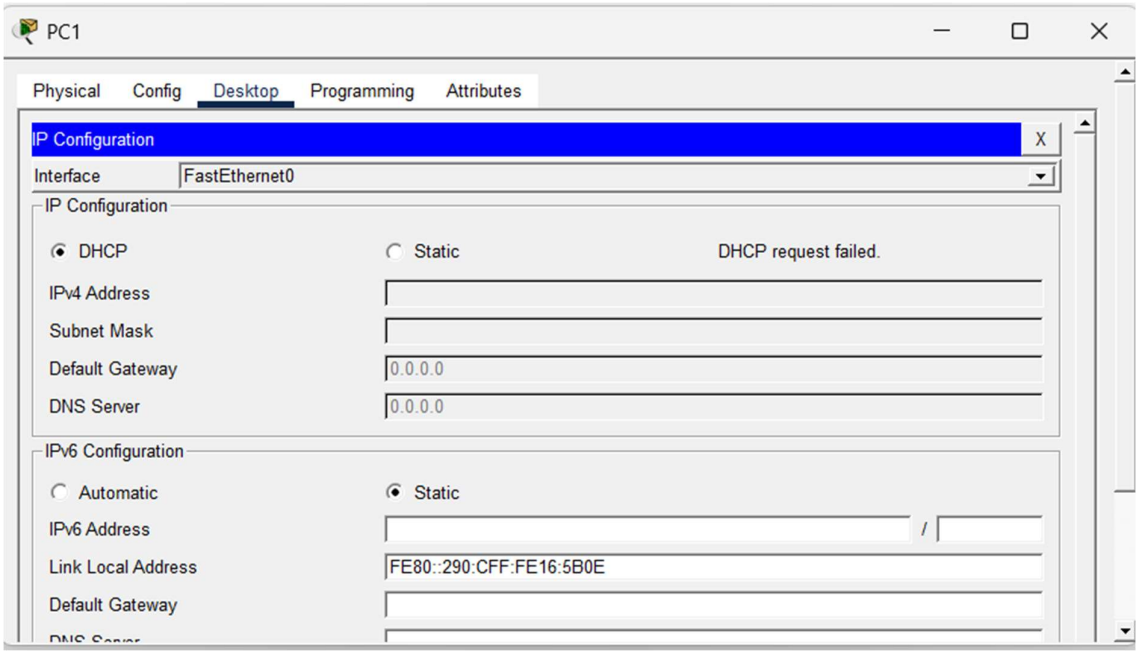


Figure 14.3.2 Failure Evidence

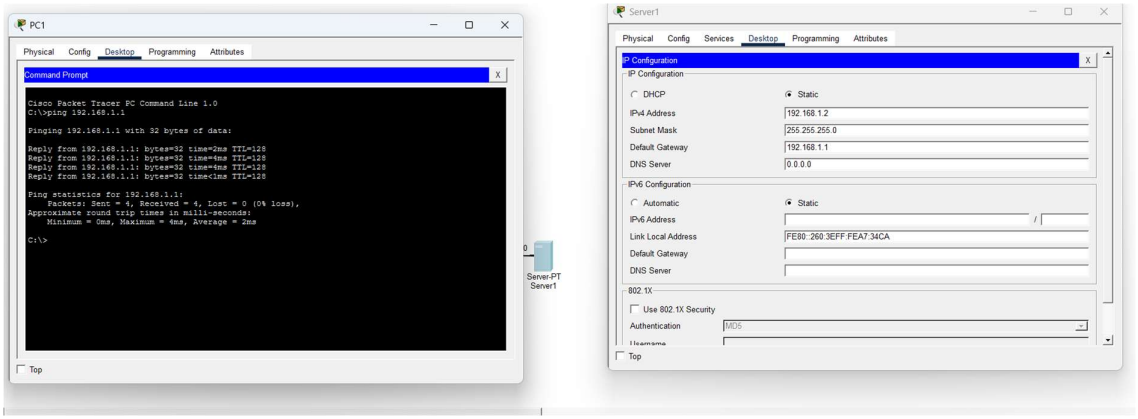


Figure 14.3.3 Successful Pinging after Fix

14.4 Test Case 04 – DNS Resolution Failure

The fourth test case focuses on DNS.

The network was configured so that a client could communicate with a server using both an IP address and a domain name.

The DNS service was initially configured correctly.

A fault was introduced by changing the DNS configuration.

As a result, the client could communicate using the server IP address but could not successfully resolve the domain name.

The network configuration was collected as evidence.

NetForensics AI analyzed the information and identified the likely DNS configuration issue.

The diagnosis was reviewed by a human.

The correct DNS configuration was restored and successful domain name resolution was verified.

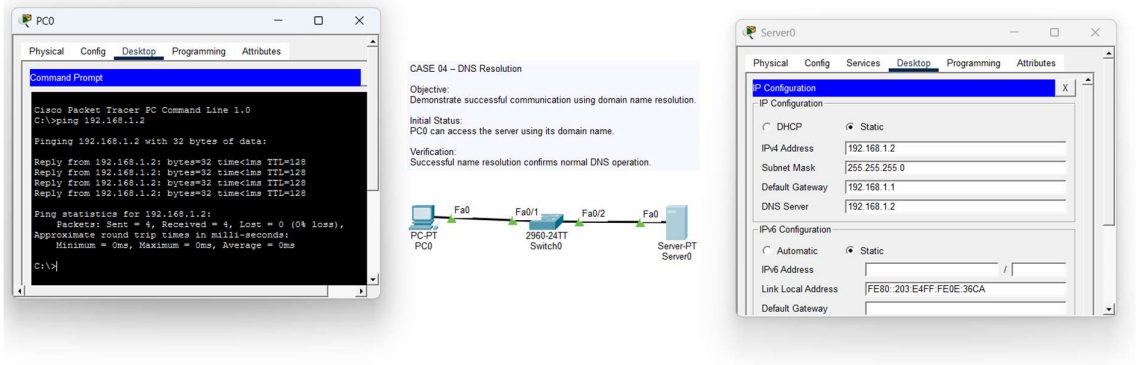


Figure 14.4.1 Working DNS Configuration

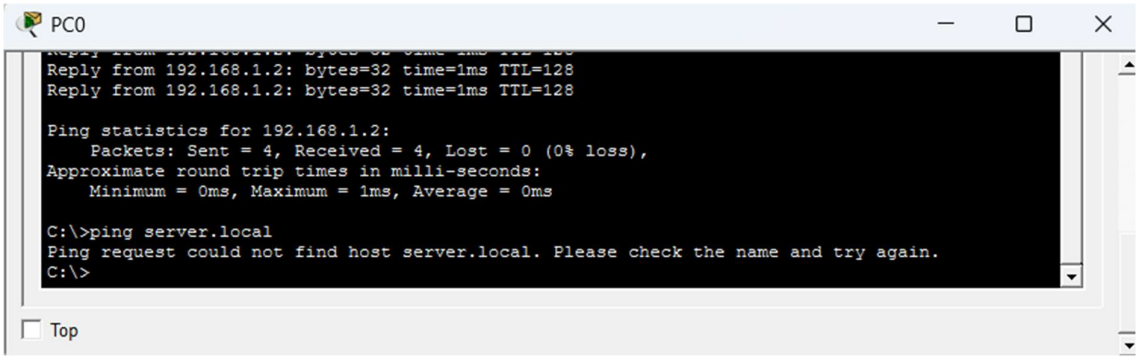


Figure 14.4.2 DNS Failure Evidence

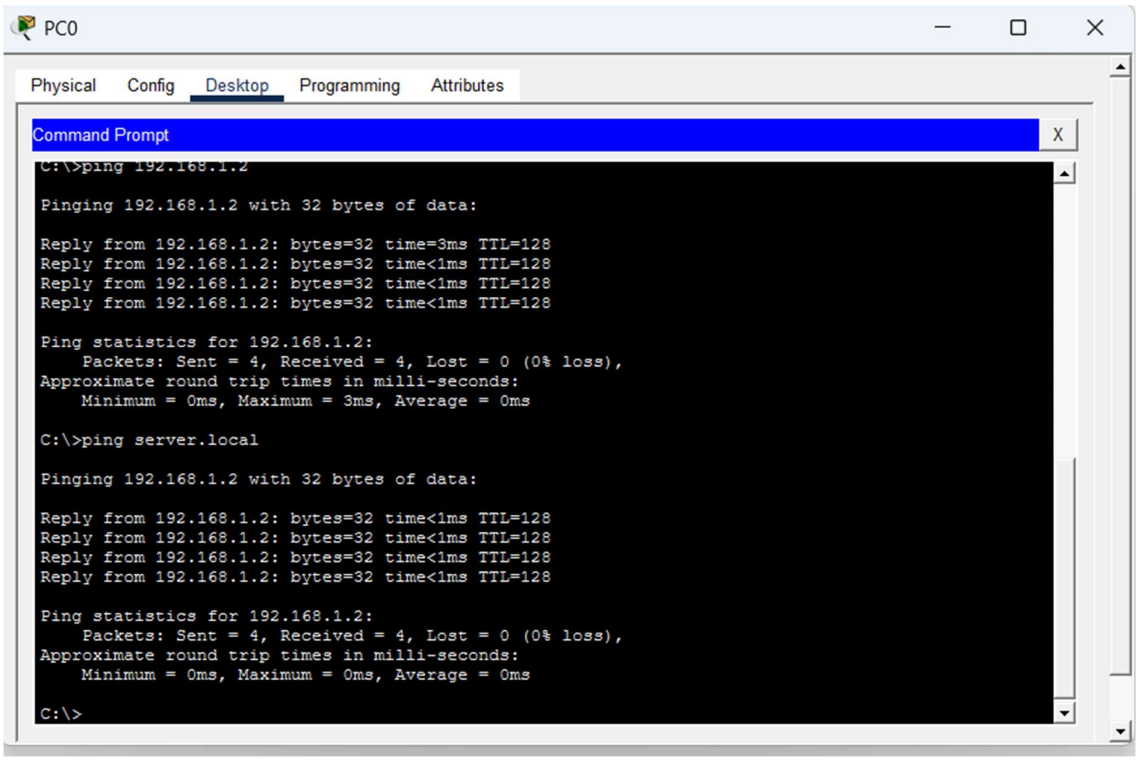


Figure 14.4.3 Successful Name Resolution After Fix

14.5 Test Case 05 – Static Routing Failure

The fifth test case demonstrates a routing configuration problem.

Two separate networks were connected using routers.

Static routes were configured to allow communication between the networks.

The working topology was verified using successful ping communication.

A fault was introduced by removing a required static route.

As a result, the source network could no longer communicate with the remote network.

The router routing table was collected as evidence.

NetForensics AI analyzed the routing information and identified the missing route.

After human review, the required static route was restored.

Connectivity between the two networks was successfully verified.

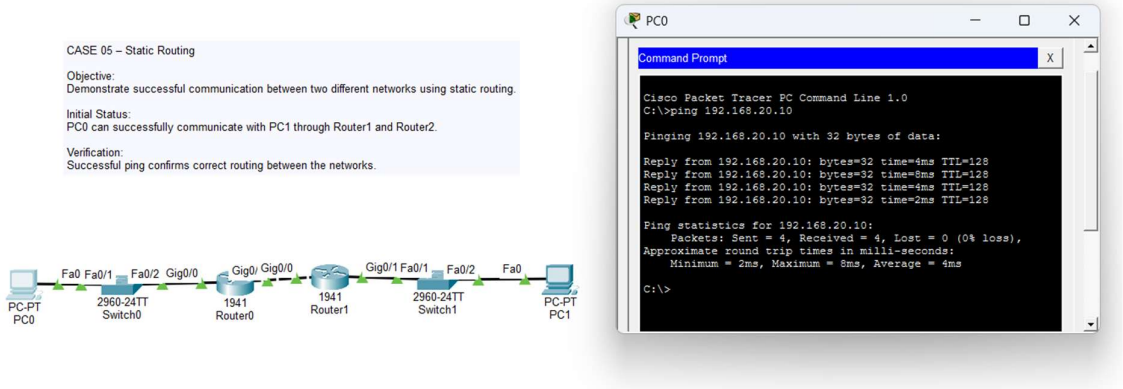


Figure 14.5.1 Working Routing Topology

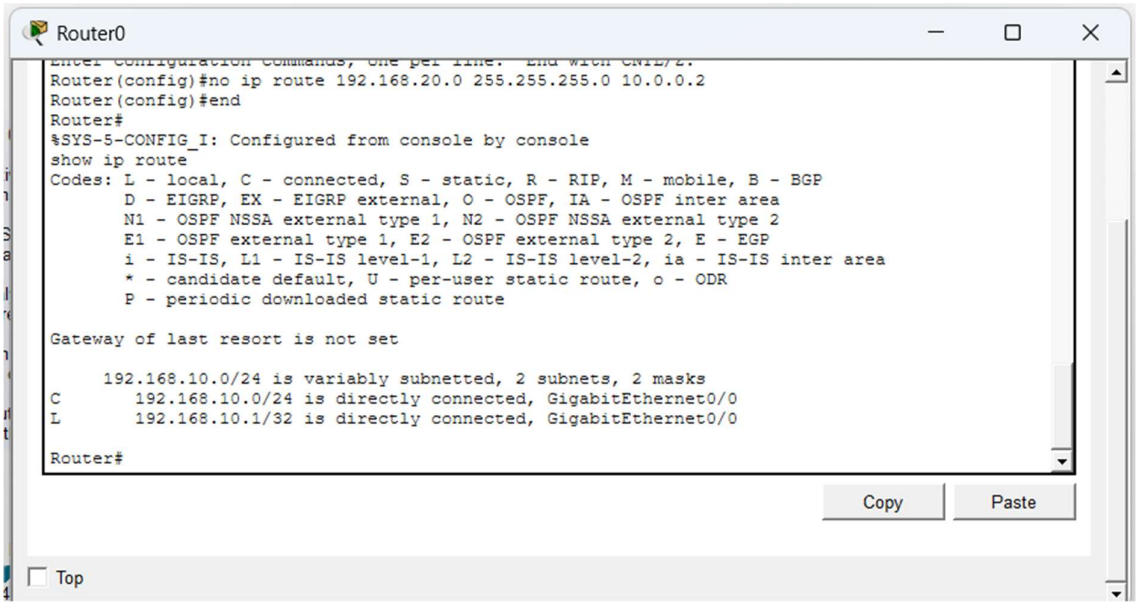


Figure 14.5.2 show ip route Evidence

15. DATA VISUALIZATION AND DASHBOARD

A dashboard was created to summarize and visualize the results of the project.

The dashboard provides an overview of network fault cases and AI-assisted diagnosis performance.

The visualizations help users understand patterns that may not be immediately visible in raw CSV files.

The dashboard includes relevant indicators such as:

- Total number of cases
- AI confidence levels
- AI match or diagnosis results
- Human review decisions
- Severity distribution
- Issue or protocol categories

Interactive filters can be used to explore specific categories.

The dashboard makes the project results easier to understand for both technical and non-technical users.

For example, a technical user may analyze the frequency of different network faults, while a non-technical evaluator can quickly understand the overall performance of the AI system through visual summaries.

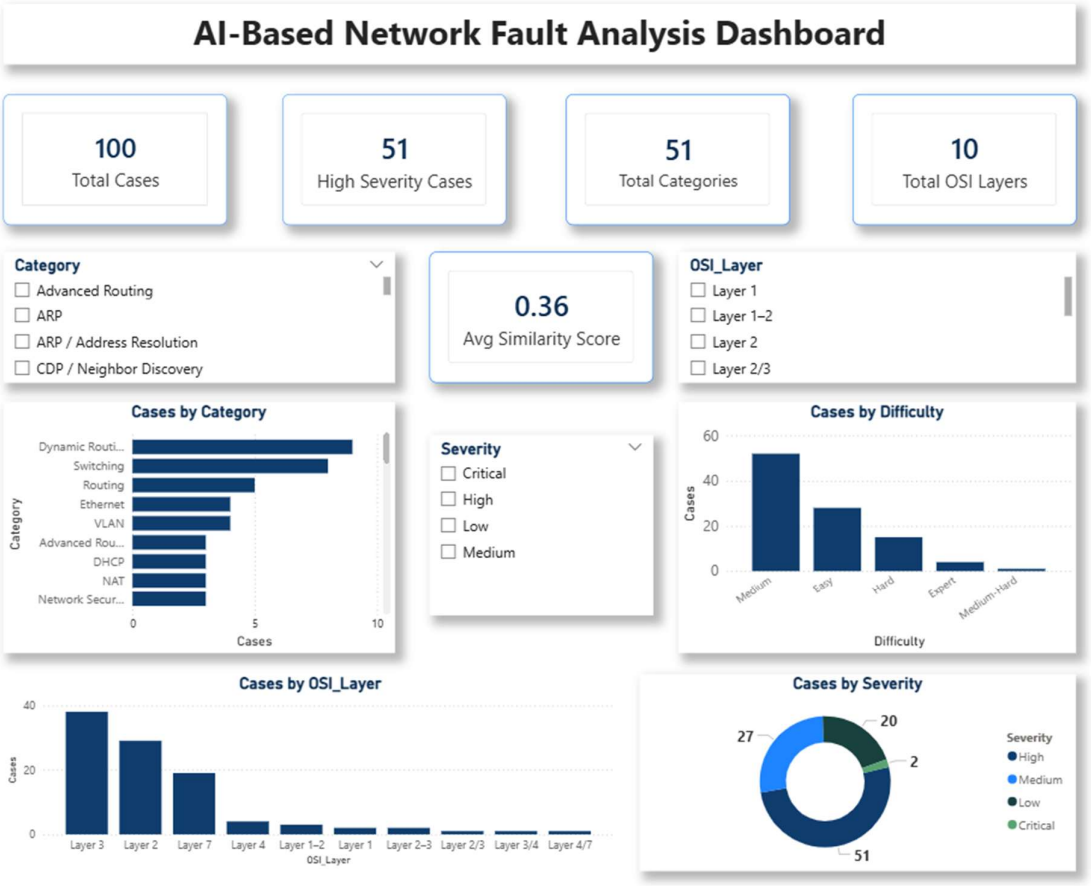


Figure 15.1 Power BI Dashboard

16. RESULTS AND DISCUSSION

The implementation of NetForensics AI demonstrates the usefulness of combining AI-assisted analysis with traditional network troubleshooting methods.

The system was able to process structured network fault information and generate recommendations including likely root causes, confidence levels, evidence references, OSI layers, next commands, and suggested fixes.

The rule-based validation component provided deterministic detection of common network configuration problems.

The human review process ensured that AI-generated responses were not automatically accepted without validation.

The Cisco Packet Tracer test cases demonstrated the practical workflow of the system.

The most important observation from the project is that the quality of an AI diagnosis depends significantly on the quality and completeness of the evidence provided.

When sufficient evidence is available, the AI can provide more confident and useful recommendations.

When evidence is incomplete, the correct approach is not to make an unsupported assumption. Instead, additional investigation and human review are required.

This finding supports the main responsible AI principle used in NetForensics AI:

AI should assist decision-making, but humans should retain control over final technical actions.

17. KEY FEATURES OF NETFORENSICS AI

The major features of the proposed system are:

1. AI-assisted network troubleshooting.
2. Structured analysis of network symptoms.
3. Analysis of Cisco-style command outputs.
4. Root cause identification.
5. AI confidence indication.
6. Evidence-based diagnosis.
7. OSI layer identification.
8. Recommended next troubleshooting commands.
9. Suggested network fix steps.
10. Rule-based validation for common configuration errors.
11. Mandatory human review.
12. Responsible AI logging.
13. Dataset-based performance analysis.
14. Interactive dashboard visualization.
15. Practical Cisco Packet Tracer test demonstrations.

18. ADVANTAGES OF THE SYSTEM

NetForensics AI provides several advantages.

Faster Initial Troubleshooting

The system can help users organize available evidence and identify possible causes more quickly.

Learning Support

Students and junior engineers can understand the relationship between network symptoms and configuration faults.

Structured Recommendations

The AI response follows a consistent format instead of providing an unstructured answer.

Evidence-Based Analysis

The system encourages the use of actual command output and network evidence.

Human Control

AI recommendations are reviewed before configuration changes are accepted.

Responsible AI Approach

The system acknowledges uncertainty and supports human correction.

Combination of AI and Rules

The project combines intelligent interpretation with deterministic validation.

Visual Analysis

The dashboard makes results easier to understand.

19. LIMITATIONS

Although the project demonstrates useful capabilities, it has certain limitations.

1. The system is primarily tested using simulated network environments.
2. AI output quality depends on the quality of evidence provided.
3. Limited command output may result in lower confidence or incomplete diagnosis.
4. The system focuses on selected common network fault categories.
5. AI recommendations still require technical validation.
6. Real-world enterprise networks may contain more complex configurations.
7. The current system does not automatically execute configuration changes.

These limitations highlight the importance of keeping a human reviewer in the troubleshooting process.

20. FUTURE SCOPE

NetForensics AI can be extended in several ways.

Integration with Real Network Devices

Future versions could analyze logs and command outputs from real routers and switches.

Automatic Log Collection

The system could automatically collect relevant troubleshooting information.

More Network Protocol Coverage

Additional protocols and technologies can be included.

Examples include:

- NAT
- ACL
- Wireless networks
- OSPF
- EIGRP
- BGP
- Network security faults

Advanced Machine Learning

Historical troubleshooting cases could be used to train machine learning models for fault classification.

Real-Time Monitoring

The system could be integrated with network monitoring tools to identify problems in real time.

Knowledge Base Integration

A larger troubleshooting knowledge base could improve recommendation quality.

Multi-Device Evidence Analysis

Future versions could automatically compare outputs from multiple devices.

Improved Dashboard

More advanced analytics and predictive insights could be added.

Automated Test Case Generation

The system could generate simulated fault scenarios for networking education and training.

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21. CONCLUSION

NetForensics AI – An Intelligent Network Fault Detection and Troubleshooting System was developed as an AI-assisted solution for analyzing common network faults.

The project addresses the challenge faced by students and junior network engineers when connecting network symptoms with their actual root causes.

The system combines several important components, including a structured network fault dataset, AI-assisted diagnosis, prompt engineering, rule-based validation, human review, responsible AI logging, visualization, and practical Cisco Packet Tracer testing.

The AI component provides structured recommendations such as likely root cause, confidence, evidence used, OSI layer, recommended next command, and suggested fix steps.

However, the project does not treat AI as an autonomous decision-maker. Human review is a mandatory part of the workflow.

The practical test cases demonstrate the complete troubleshooting process from working network creation to fault generation, evidence collection, AI analysis, human review, configuration correction, and final verification.

The dashboard provides a visual summary of the collected results and helps users understand patterns in network fault cases and AI performance.

The project demonstrates that Artificial Intelligence can be effectively used as an intelligent assistant in network troubleshooting. The most important contribution of NetForensics AI is the combination of AI assistance with evidence-based reasoning and responsible human oversight.

Therefore, NetForensics AI provides a foundation for future intelligent network troubleshooting systems that are not only automated and efficient but also transparent, explainable, and responsible.

REFERENCES

1. Cisco Networking Academy – Networking Fundamentals and Cisco Networking Concepts.
2. Cisco Packet Tracer Documentation.
3. Python Documentation.
4. Pandas Documentation.
5. Jupyter Notebook Documentation.
6. Microsoft Power BI Documentation.
7. OSI Reference Model and Computer Networking Concepts.
8. Relevant Artificial Intelligence and Responsible AI learning resources.

APPENDIX

Appendix A – Dataset Columns

Include screenshots or descriptions of the dataset structure.

Appendix B – AI Prompt Templates

Include the main diagnosis prompt and supporting prompt templates.

Appendix C – Rule-Based Detection Output

Include sample outputs from the Python rule checker.

Appendix D – Human Review Log

Include examples of Accepted, Edited, and Rejected AI diagnoses.

Appendix E – Cisco Packet Tracer Test Cases

Include additional screenshots for:

- Working topology
- Fault generation
- Evidence collection
- AI diagnosis
- Applied fix
- Final verification

Appendix F – Dashboard

Include the complete dashboard screenshot and key visual insights.

Appendix G – Supporting Course Certificates

The following learning courses were completed as part of the learning and skill development associated with this project:

1. Data Analytics Essentials
2. Analyze Customer Reviews
3. Introduction to Modern AI

The knowledge gained from these courses supported the development of the project in areas including data analysis, artificial intelligence concepts, AI-assisted decision-making, and analytical workflows.

The corresponding course completion certificates are included separately in the project submission folder under the “Certificates” directory.